1) VISUALIZATION & PRE-PROCESSING IMAGES

pip install tensorflow numpy matplotlib opencv-python scikit-learn

Requirement already satisfied: tensorflow in /usr/local/lib/python3.11/dist-packages (2.18.0)
Requirement already satisfied: numpy in /usr/local/lib/python3.11/dist-packages (2.0.2)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.11/dist-packages (3.10.0)
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mdurl~=0.1 in /usr/local/lib/python3.11/dist-packages (from markdown-it-py>=2.2.0->rich->keras>=3.5.0
```

Mount path

```
from google.colab import drive

# Mount your Google Drive
drive.mount('/content/drive', force_remount=True)
```

```
Mounted at /content/drive
```

```
import os

# Check your Google Drive contents
drive_path = '/content/drive/My Drive/' # Mounted drive root directory

# List contents of the drive root or specific folders to find your directories
os.listdir(drive_path) # This will list top-level contents
```

```
['2)Bioinformatics_work',
 'my_projects',
 '1)Molecular_collection',
 'phd_preparation',
 'Bulk cell transcriptomic data analysis.gdoc',
 'nr14_old_drive',
 'COLLECTIONS',
 'Colab Notebooks',
 'my_work',
 'PYTHON']
```

Load data

```
adenocarcinoma_traning = '/content/drive/My
Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/data/training/Adenocarcin
normal_training = '/content/drive/My
Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/data/training/Normal_training'

adenocarcinoma_test = '/content/drive/My
Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/data/test/adenocarcinoma_tes
normal_test = '/content/drive/My
```

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Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/data/test/norma_test'

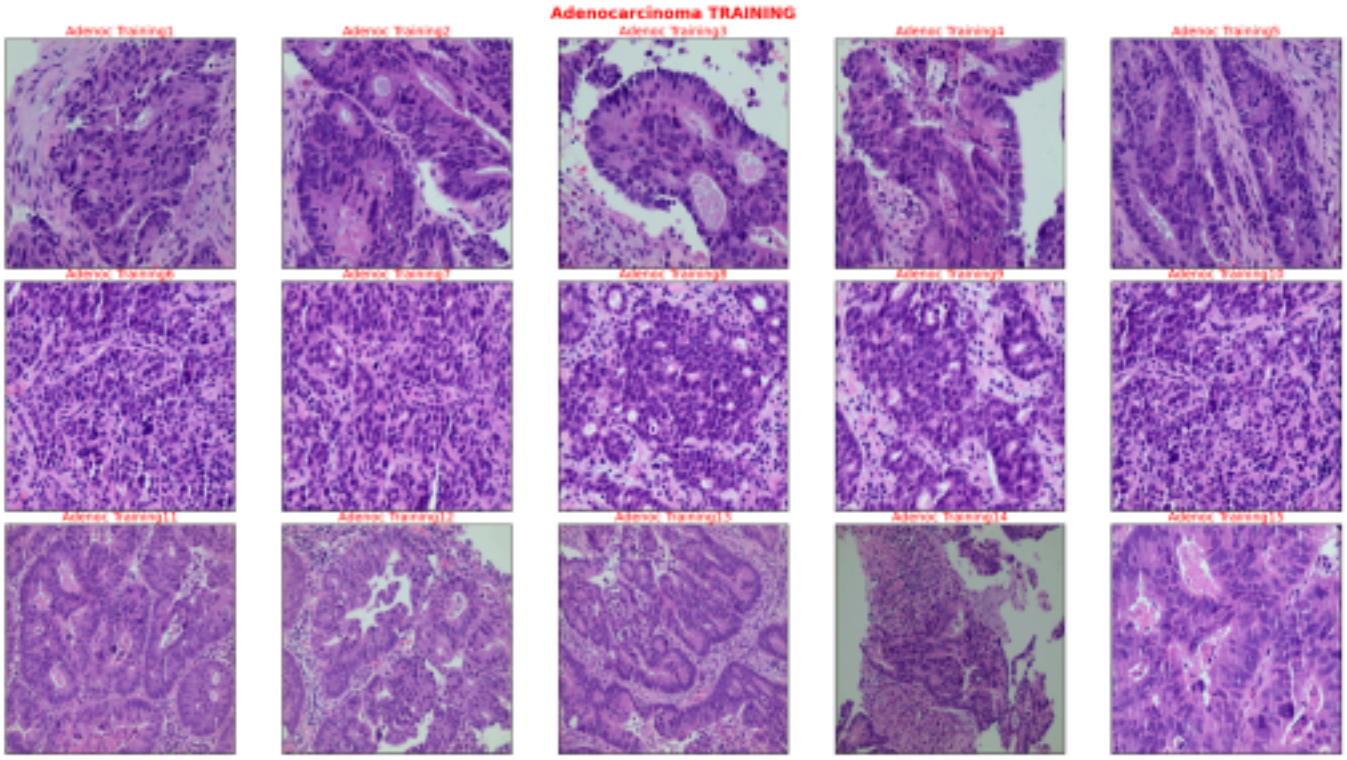
Visualization

1_adenocarcinoma_training:

```
import os
import matplotlib.pyplot as plt

# Adenocarcinoma TRAINING
adenocarcinoma_traning = '/content/drive/My
Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/data/training/Adenocarcin
plt.figure(figsize=(22, 12))

for i, f in enumerate(sorted(os.listdir(adenocarcinoma_traning))[:15]):
    plt.subplot(3, 5, i+1, xticks=[], yticks=[])
    img = plt.imread(os.path.join(adenocarcinoma_traning, f))
    plt.imshow(img)
    plt.title(f"Adenoc Training{i+1}", fontsize=14, pad=4, color='red') # Red title
plt.suptitle("Adenocarcinoma TRAINING", y=0.98, fontsize=18, color='red',
weight='bold') # Red main title plt.tight_layout(w_pad=0.02, h_pad=0.04)
plt.show()
```

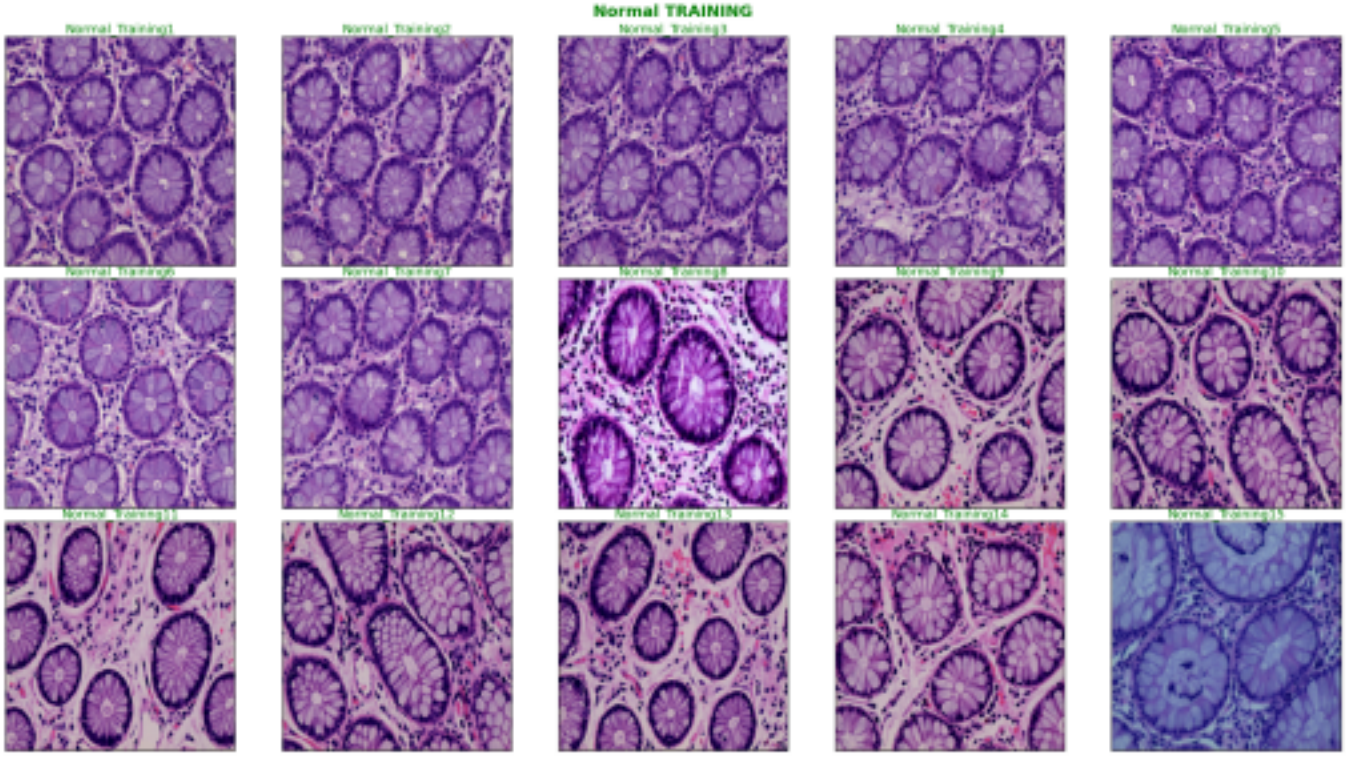


2_Normal_TRAINING:

```
# Normal TRAINING
normal_training = '/content/drive/My
Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/data/training/Normal_training'
plt.figure(figsize=(22, 12))

for i, f in enumerate(sorted(os.listdir(normal_training))[:15]):
```

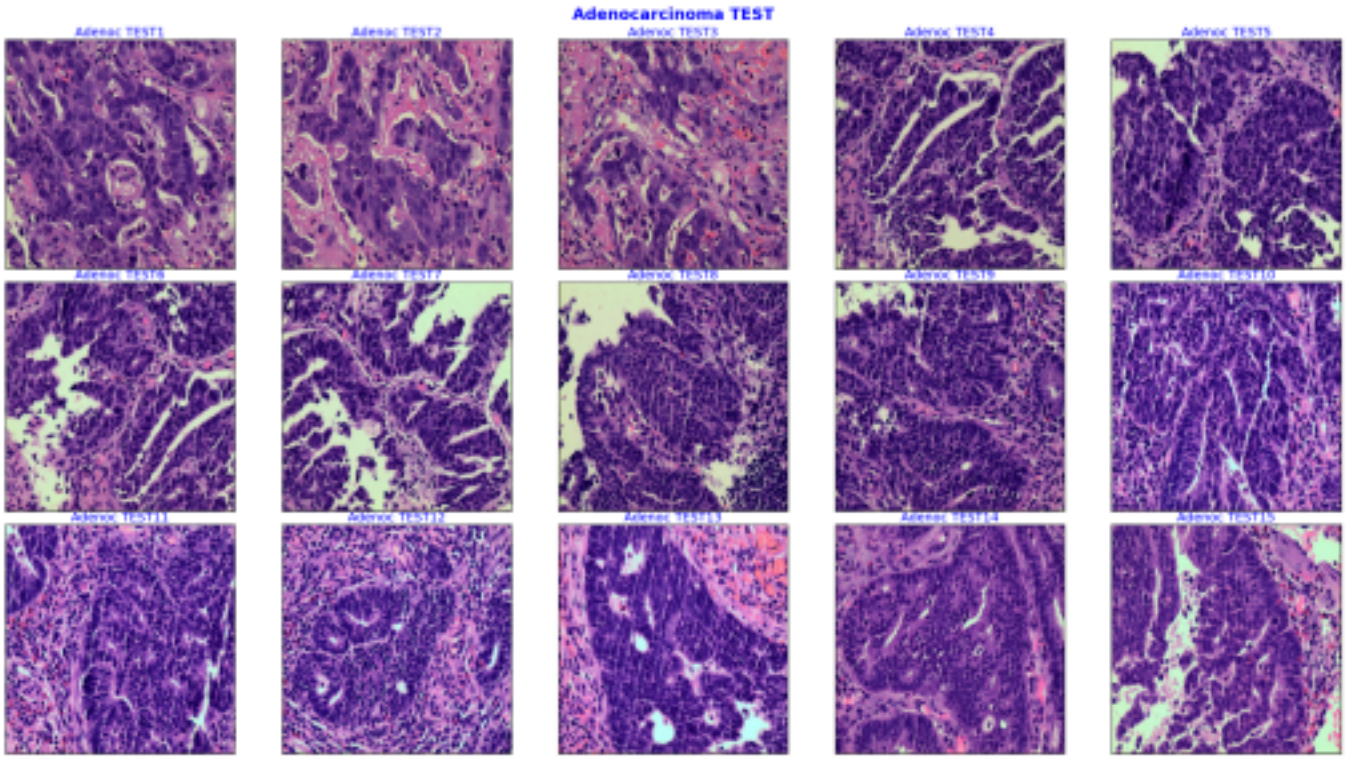
```
    plt.subplot(3, 5, i+1, xticks=[], yticks=[])
    img = plt.imread(os.path.join(normal_training, f))
    plt.imshow(img)
    plt.title(f"Normal_Training{i+1}", fontsize=14, pad=4, color='green')
plt.suptitle("Normal TRAINING", y=0.98, fontsize=18, color='green', weight='bold')
plt.tight_layout(w_pad=0.02, h_pad=0.04)
plt.show()
```



3_Adenocarcinoma_TEST

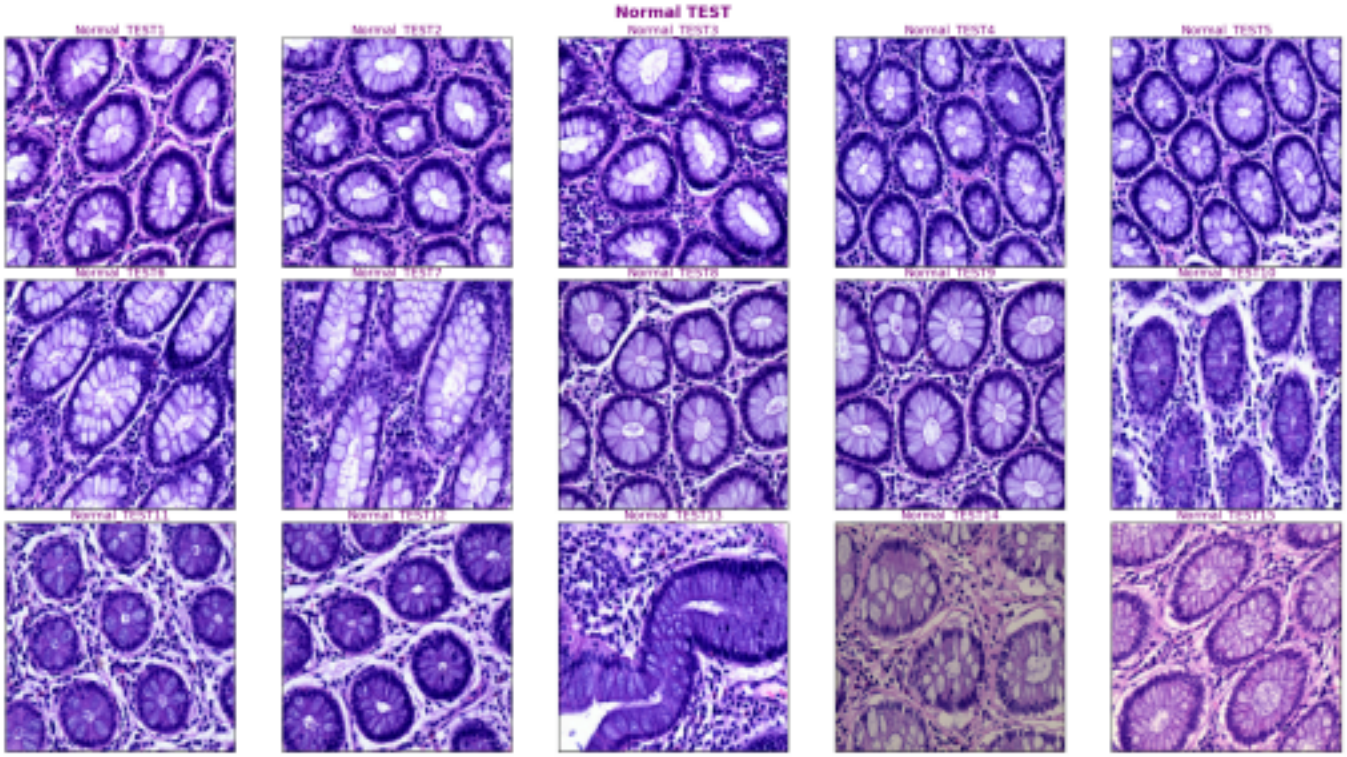
```
# Adenocarcinoma TEST
adenocarcinoma_test = '/content/drive/My
Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/data/test/adenocarcinoma_tes
plt.figure(figsize=(22, 12))

for i, f in enumerate(sorted(os.listdir(adenocarcinoma_test))[:15]):
    plt.subplot(3, 5, i+1, xticks=[], yticks=[])
    img = plt.imread(os.path.join(adenocarcinoma_test, f))
    plt.imshow(img)
    plt.title(f"Adenoc TEST{i+1}", fontsize=14, pad=4, color='blue')
plt.suptitle("Adenocarcinoma TEST", y=0.98, fontsize=18, color='blue', weight='bold')
plt.tight_layout(w_pad=0.02, h_pad=0.04)
plt.show()
```




```
# Normal TEST
normal_test = '/content/drive/My
Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/data/test/norma_test'
plt.figure(figsize=(22, 12))

for i, f in enumerate(sorted(os.listdir(normal_test))[:15]):
    plt.subplot(3, 5, i+1, xticks=[], yticks=[])
    img = plt.imread(os.path.join(normal_test, f))
    plt.imshow(img)
    plt.title(f"Normal_TEST{i+1}", fontsize=14, pad=4, color='purple')
plt.suptitle("Normal TEST", y=0.98, fontsize=18, color='purple', weight='bold')
plt.tight_layout(w_pad=0.02, h_pad=0.04)
plt.show()
```



Data pre-process

```
import os
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

Define preprocessing functions

```
import cv2
import os
import numpy as np
import matplotlib.pyplot as plt

IMG_SIZE = 128 # Standard size for resizing

# Function to resize images
def resize_image(img, size=IMG_SIZE):
    return cv2.resize(img, (size, size))

# Function to apply Gaussian blur (for noise reduction)
def apply_gaussian_blur(img):
    return cv2.GaussianBlur(img, (3, 3), 0)

# Function to apply Contrast Enhancement (CLAHE in LAB color space)
def enhance_contrast_color(img):
    clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8,8))
    lab = cv2.cvtColor(img, cv2.COLOR_BGR2LAB)
    l, a, b = cv2.split(lab)
    l = clahe.apply(l)
    lab = cv2.merge((l, a, b))
    return cv2.cvtColor(lab, cv2.COLOR_LAB2BGR)
```

```
# Function to apply Edge Detection overlay
def apply_edge_overlay(img, alpha=0.6):
    edges = cv2.Canny(img, 50, 150) # Detect edges
    edges_colored = cv2.cvtColor(edges, cv2.COLOR_GRAY2BGR) # Convert to 3-channel
    return cv2.addWeighted(img, alpha, edges_colored, 1 - alpha, 0) # Blend images


# Function to preprocess ALL images in a folder
def load_and_preprocess_all(folder):
    processed_images = []
    filenames = sorted(os.listdir(folder)) # Get all image filenames

    for filename in filenames:
        img_path = os.path.join(folder, filename)
        img = cv2.imread(img_path) # Load in color
        if img is not None:
            img_resized = resize_image(img)
            img_blur = apply_gaussian_blur(img_resized)
            img_contrast = enhance_contrast_color(img_blur)
            img_edge_overlay = apply_edge_overlay(img_contrast) # Edge overlay
            processed_images.append((filename, img_edge_overlay)) # Store all images
    return processed_images
```

PRE-PROCESS adenocarcinoma Training images

Save all processed images

```
# Define input folders
input_folders = {
    "adenocarcinoma_traning":                                                    "/content/drive/My
    Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/data/training/Adenoc
    "normal_training":                                                           "/content/drive/My
    Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/data/training/Normal_traini
    "adenocarcinoma_test":                                                       "/content/drive/My
    Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/data/test/adenocarcinom
    "normal_test":                                                               "/content/drive/My
    Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/data/test/norma_test"
}


# Define output folders (Keys must exactly match input_folders)
output_folders = {
    "adenocarcinoma_traning": "/content/drive/My
Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/preprocessed/prep_ad
"normal_training": "/content/drive/My
Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/preprocessed/prep_normal_tr
"adenocarcinoma_test": "/content/drive/My
Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/preprocessed/prep_adeno
"normal_test": "/content/drive/My
Drive/PYTHON/ML_image_process/binary_classification_adenocarcinoma/preprocessed/prep_normal_test/"
}
```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